Missing citations in "A spectral gap throughout the high-temperature Sherrington–Kirkpatrick phase"

October 13, 2026

Summary of findings

Each row is a citation the preprint is missing: earlier work that already contains part of the preprint, which the preprint does not cite where that part appears. "Cited?" names the earlier work and says what, if anything, the preprint cites it for. The exception is a row flagged *overstated novelty*: there the preprint cites the work, but its framing presents the content as new. Terms are defined in Section 1; centrality is a best-effort judgment. Locations use the preprint's own numbering and printed pages.

ID	Relationship	Centrality	Technique or result	In the preprint	Cited?
F1	*same technique*	central	Planted law of disorder and observation path, likelihood ratio $Z/\mathbb{E}Z$, and transfer back to the original disorder	(5.4) (p. 28); size bias, (5.27) and transfer (pp. 39–40)	El Alaoui–Montanari–Sellke 2022 [1]: credited elsewhere (named in §1.1, p. 4, not in §5); Aizenman–Lebowitz–Ruelle 1987 [2]: credited elsewhere (p. 3)
F2	*adapted*	central	Stopping the localization before its path estimates can fail, with early stops and terminal failures as one exceptional event	§2.2, (2.3)–(2.4) (pp. 7–8); §2.4 (p. 10); Cor. 5.2 (p. 27)	Huang–Mohanty–Rajaraman–Wu 2024 [3]: not cited; Chen–Eldan 2022 [4]: credited in part; **attribution gap**
F3	*adapted*	central	Deterministic equivalents for the diagonal of words with an Onsager-shifted inverse, uniform over diagonal parameters chosen after the matrix	Lemmas 3.1 and 3.3, (3.18) (pp. 12–19)	Jekel–Sandhu–Shi 2024 [5]; Davies–Lee–Sandhu–Shi 2026 (PHA III) [6]; Fan–Mei–Montanari 2018 [7]: not cited; Speicher 1998 [8]: credited in part; **attribution gap**
F4	*adapted*	central	Planted Kac–Rice first moment for TAP critical points: determinant rate, Gaussian density of the TAP map, complexity functional and its sign	Lemma 3.5 (p. 20); Lemma 4.1 (pp. 22–24); (5.5), Lemma 5.3, (5.9), Lemma 5.4 (pp. 28–31)	Fan–Mei–Montanari 2018 [7]: not cited; **attribution gap**

ID	Relationship	Centrality	Technique or result	In the preprint	Cited?
F5	*adapted*	central	Control of approximate critical points from an annealed first moment: smoothed integral, local-volume lower bound, determinant cancellation	(5.5) (p. 28); Lemma 5.6 (pp. 35–37); exclusion (pp. 37–38); Lemma 3.5 (pp. 20–21)	Huang–Sellke 2023 [9], Huang–Montanari–Pham 2024 [10]: not cited; **attribution gap**
F6	*adapted*	central	Well-posedness of the TAP equation along the localization path: planted law, first-moment exclusion with a conditional Hessian, approximate critical points, transfer	Prop. 5.1 (p. 27); §5.4 (pp. 37–40); Lemma 5.7 (pp. 40–41)	Huang–Montanari–Pham 2024 [10]: not cited; **attribution gap**
F7	*adapted*	central	TAP Hessian conditioned on a vanishing gradient, positive near the Bayes law, inside a Kac–Rice first moment	(5.8) (pp. 28–29); Lemma 5.5 (pp. 31–35); (5.25) (p. 37)	Celentano–Fan–Mei 2021 [11]: not cited; **attribution gap**
F8	*adapted*	central	A memory correction for every input of a multi-input recursion, and its cancellation of one-step reversals at every depth	Def. 6.2, (6.7) (p. 44); (6.14) (p. 46); proof of Lemma 6.5 (p. 54); proofs of Lemmas 6.1, 6.4 and 3.3 (pp. 43, 51–52, 16)	Bayati–Lelarge–Montanari 2012 [12]; Montanari 2018 [13]: not cited (Thouless–Anderson–Palmer cited for the Onsager correction, Bolthausen for (6.1); AMP named only in §1.1, p. 4, citing others for other content); **attribution gap**
F9	*adapted*	central	Gap at a large observation time from few weak-field coordinates and uniformly small principal submatrices	Lemma 8.2, Prop. 8.3 (pp. 61–64)	Davies–Lee–Sandhu–Shi 2026 (PHA III) [6], Davies–Lee–Sandhu–Shi 2026 (weak Poincaré) [14]: not cited; Bandeira–El Alaoui–Rödder 2026 [15]: credited in part; **attribution gap**
F10	*same route*	central	Poincaré inequality for all $\beta < 1$ from operator-norm covariance bounds at the localization fields, via a uniformly positive TAP Hessian	§1.4 (p. 6); Props. 5.1, 7.2; §2	El Alaoui–Gaitonde 2022 [16]: cited for other content (Thm 1.1); Brennecke–Xu–Yau 2023 [17]: not cited
F11	*weaker result*	central	Weak Poincaré inequality for zero-field SK heat-bath dynamics at $\beta < 1/2$	Abstract; Thm 1.1 (p. 3)	Davies–Lee–Sandhu–Shi 2026 (weak Poincaré) [14]: not cited

ID	Relationship	Centrality	Technique or result	In the preprint	Cited?
F12	*same technique*	supporting	Margin for the Onsager-shifted TAP Hessian by comparing $v^{\mathsf{T}}Wv$ with a norm-weighted linear process under a common shift	Lemma 3.2, (3.8)–(3.9) (pp. 12–13)	Celentano–Fan–Mei 2021 [11]: not cited; Celentano 2022 [18]: cited for other content; **attribution gap**
F13	*same technique*	supporting	Consistent Gaussian law: mean equals variance on the Nishimori line, then a one-variable fixed point with uniqueness, size and Lipschitz bounds	Lemma 4.2, (4.7)–(4.11) (pp. 25–26)	El Alaoui–Montanari–Sellke 2022 [1]: cited for other content; Davies–Lee–Sandhu–Shi 2026 (PHA III) [6]: not cited; **attribution gap**
F14	*fragment*	supporting	Path-law entropy of the reweighted observations as the entropy lost along the localization	§2.3, (2.5) and (2.7) (pp. 8–9)	Chen–Eldan 2022 [4]: cited for other content (Lehec [19] is cited for the identity)
F15	*fragment*	supporting	Stein operator of the cosh-tilted Gaussian mixture and its bounded solution for tanh	(6.8)–(6.9) (p. 45), used in Lemma 6.5 (pp. 53–54); (6.12)–(6.13) (pp. 45–46); (7.3) (p. 57)	Chatterjee 2007 [20]: not cited
F16	*fragment*	supporting	Comparing a posterior mean with the TAP root through a late Onsager-corrected iterate	(6.10)–(6.11) (p. 45); Prop. 7.3 (p. 58)	El Alaoui–Montanari–Sellke 2022 [1]: credited elsewhere (§1.1, p. 4); Celentano–Fan–Mei 2021 [11]: not cited

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1 About this document

The preprint. The preprint is *A spectral gap throughout the high-temperature Sherrington–Kirkpatrick phase*, dated September 24, 2026, with author line "OpenAI" [21]. It is one of the manuscripts of the openai/math release. It proves that for every fixed inverse temperature $0 < \beta < 1$, with probability tending to one over the disorder, the zero-field Gaussian Sherrington–Kirkpatrick Gibbs measure satisfies a Poincaré inequality for the unscaled single-site heat-bath Dirichlet form, with a constant depending only on β; equivalently, the heat-bath chain has spectral gap of order $1/n$. Section 2 summarizes the paper and fixes the notation used throughout this document. The preprint cites no other preprint from the same release.

What this document reports. Each finding is a citation the preprint is missing: part of the preprint's mathematics (a technique, construction, estimate, proof route or result) that an identifiable earlier paper already contains, where the preprint does not cite that paper for it where it is used. "Earlier" means publicly available before the preprint's date. The one exception, overstated novelty, is described below. Not reported: textbook material in its textbook form; weaker versions of intermediate statements; and, for content the preprint uses, cases where it cites a later source that covers that content rather than the original. An earlier result on the same problem is reported even when later work that the preprint cites implies it. Findings are ordered by centrality, then by relationship in the order listed below, then by position in the preprint.

Relationships. A finding's relationship is the one between the preprint and the first earlier work listed for it (the principal), the work with the highest relationship in the order below; when several hold that relationship, however closely each matches, the one the others credit or build on, otherwise

the earliest. Any other works listed relate to the same content more weakly or in part; the finding says how, including which work originated the idea when that is not the first.

same result. An earlier paper, or several earlier papers together, prove essentially the same statement.

same technique. The earlier technique, construction or theory is reused essentially intact, possibly in a new setting or with new ingredients around it.

adapted. The earlier technique's structure is followed, with a change at its core: a different object, or a key step replaced.

proof shortcut. A published result would replace or substantially shorten a large part of the argument.

same route. An earlier paper proposed the proof strategy, even heuristically, and the preprint carries it out.

fragment. One identifiable piece of the earlier technique is reused.

weaker result. An earlier paper proves a weaker, special-case or partial-range version of one of the preprint's main results, or of a statement it presents as a significant contribution.

Centrality. Each finding also gets a best-effort judgment of how much of the preprint's main argument depends on it: *main result* (reserved for earlier work that covers the main result itself, or most of it), *central*, *supporting* or *peripheral*. An earlier weaker result is never labelled main result. Each finding states the chain of results it feeds, so the basis for the judgment is visible.

Citations. The "Cited?" column of the summary names the earlier work and says what the preprint cites for the content in question: *not cited* (the work is not in the bibliography, and nothing the preprint cites credits it for this content), *cited for other content*, *credited elsewhere* (credited for this content, directly or through a cited location that credits it, but not where it is used, in a way that an expert reader who doesn't know the earlier work might not connect with it), or *credited in part*. A citation counts only when it is *for that specific content*; citing the same paper nearby for a different result does not count. Content the preprint credits where it is used, directly or through a cited location that itself credits the earlier work, is not reported, except when its novelty is overstated (below). For overstated novelty, the column says where the preprint does cite the work.

Flags.

Attribution gap. Marks the most serious missing citations: the relationship is same technique, adapted or same result; the centrality is main result, central or supporting; and the preprint cites the principal earlier work for this content nowhere.

Overstated novelty. The preprint does cite the earlier work for this content, but its abstract, a statement of its significant contributions, or the framing of a main result presents that content as new, explicitly or in the way a reader would reasonably take it. This is the one case in which content the preprint credits is reported.

How to check a claim. Locations in the preprint use its section, equation and theorem numbers and the page numbers printed in its PDF. Locations in other sources give the theorem, equation and printed page of the version listed in the bibliography. Each finding ends with a short checklist of the passages to put side by side.

Provenance. This document was generated by LLM agents (Claude Opus 5.5) running the precedence-review skill (jnshi/math-lineage, commit `e025345ea`), makes no claim of completeness, and has not been vetted by a human. The run was used to develop the skill, and its later stages ran revisions of the skill made during the run, the last of them commit `e1e8ffbf7`. It was run by Jonathan Shi. The leads for this run came from the findings of an earlier run with version 1 of the skill, in which Jonathan Shi supplied the lead that became finding F3. Jonathan Shi makes no claim of impartiality.

2 The preprint in brief, and notation

Model and theorem. Fix $0 < \beta < 1$ and put $j = \beta^2$. The disorder J is a real symmetric $n \times n$ matrix with zero diagonal and independent entries $J_{ij} \sim N(0, j/n)$, $i < j$. For a field $h \in \mathbb{R}^n$,

$$\mu_h(x) = \frac{1}{Z_h} \exp\Big\{ \tfrac{1}{2} x^{\mathsf{T}} J x + h^{\mathsf{T}} x \Big\}, \qquad x \in \{-1, 1\}^n,$$

and μ_0 is the zero-field Gibbs measure. With P_i^h the conditional expectation under μ_h given all spins but the ith, the unscaled heat-bath Dirichlet form is $\mathcal{D}_h(f) = \sum_i \mathbb{E}_{\mu_h}(f - P_i^h f)^2$ (1.1), p. 3. The main result, Theorem 1.1 (p. 3), states that there is $C_\beta < \infty$ with

$$\mathbb{P}_J\big\{ \mathrm{Var}_{\mu_0}(f) \le C_\beta \, \mathcal{D}_0(f) \text{ for every } f : \{-1, 1\}^n \to \mathbb{R} \big\} \longrightarrow 1,$$

so the discrete-time heat-bath chain that updates a uniform site has spectral gap at least $1/(C_\beta n)$ with probability tending to one.

The proof, in the preprint's own terms. The preprint describes two parts: "prove useful inequalities for posterior Gibbs laws, then transfer them to the unobserved law" (§1.3, p. 5). It observes a spin sample in Gaussian noise,

$$X \sim \mu_0, \qquad Y(t) = tX + B(t), \quad 0 \le t \le T,$$

with B a standard Brownian motion, as in (1.4), p. 6, so that the posterior spin law at time t is $\mu_{Y(t)}$. The transfer is proved in the preprint's §2 from three inputs (§1.4, p. 6):

1. stability of every approximate solution of the field equation along ordinary observation paths, except on an event of probability e^{-an} (Proposition 5.1, p. 27), proved in §5 with random-matrix estimates from §3 and a scalar entropy inequality from §4;
2. two estimates at such a good field h, proved in §7 from the residual identities of §6: the directional covariance bound

   $$\|\mathrm{Cov}_{\mu_h}(G, x)\|^2 \le C\big(\mathrm{Var}_{\mu_h}(G) + \mathcal{D}_h(G)\big)$$

   of (1.5), p. 6 and Proposition 7.2, p. 56, and the comparison of the mean of $\nu = f^2 \mu_h / \mathbb{E}_{\mu_h} f^2$ with the root in (1.6), p. 6 and Proposition 7.3, p. 58;
3. a terminal gap $\mathrm{Var}_{\mu_{Y(T)}} f \le 2\mathcal{D}_{Y(T)}(f)$ at a large fixed time T (Proposition 8.3, p. 63), proved in §8.

Field equation and stability. For $y \in \mathbb{R}^n$ put $m(y) = \tanh y$, $q(y) = n^{-1}\sum_i m_i(y)^2$, $b(y) = 1 - q(y)$ and $V_y = \operatorname{diag}(1 - m(y)^2)$. The field map of (5.1), p. 26 is

$$F_{J,h_0}(y) = y - h_0 + \big(jb(y)I - J\big)m(y),$$

and for a nonnegative diagonal matrix A the stability matrix of (5.2), p. 27 is

$$\mathsf{H}_J(y, A) = I + A^{1/2}\Big(jb(y)I - J - \frac{2j}{n}m(y)m(y)^{\mathsf{T}}\Big)A^{1/2},$$

the symmetric version of the derivative $F'_{J,h_0}(y)$ when $A = V_y$. At the fields under consideration, $r(h)$ denotes the unique zero of $F_{J,h}$ and $t_*(h) = \tanh r(h)$ (§1.4, p. 6).

Matrix notation of §3. The preprint couples J to a centered GOE matrix W with $\mathbb{E}W_{ab}^2 = j/n$ $(a < b)$ and $\mathbb{E}W_{aa}^2 = 2j/n$ through $J = W - \operatorname{diag}(W)$. For a nonnegative diagonal A with entries at most $1 + \eta$, where $j(1 + \eta)^2 < 1$, it sets $\chi_A = n^{-1}\operatorname{Tr} A$ and, in (3.2), p. 10,

$$K_A(z; M) = (I - zAM + z^2 j\chi_A A)^{-1}, \qquad S_A(z; M) = I + z^2 j\chi_A A - zA^{1/2}MA^{1/2}.$$

The recursion of §6. Started from a Gibbs sample x at a field h_0, the primary recursion of (6.1), p. 42 is

$$m^0 = x, \quad b_0 = 0, \quad m^l = \tanh h^l, \quad b_l = \frac{1}{n}\sum_i (1 - (m_i^l)^2), \quad h^{l+1} = h_0 + Jm^l - jb_l m^{l-1},$$

with $h^1 = h_0 + Jx$ and residuals $R_l = m^{l-1} - m^l$.

3 Findings

F1. A planted law for the disorder and the observation path, transferred back through the normalized partition function

Summary. El Alaoui, Montanari and Sellke analyse their Sherrington–Kirkpatrick sampler under a planted law of disorder and observation path, prove that its likelihood ratio against the original law is the normalized partition function, and transfer back by contiguity [1, §4.1, Prop. 4.2, pp. 10–12]. The preprint names that paper only in its introduction (§1.1, p. 4), for combining "Gaussian observations, planted laws, and approximate message passing"; in its §5, where it introduces its planted law (5.4), p. 28 and derives the size bias and the transfer back (pp. 39–40), it cites nothing. The shared approach: the computations along the path need an explicit law, and size-biasing the law of disorder, spin sample and path by the normalized partition function gives one (uniform spins, a GOE matrix plus a rank-one spike, path $tx+B$); as that ratio is not exponentially small for $\beta < 1$, failure probabilities that vanish under the planted law (exponentially fast, in the preprint's form) also vanish under the original one, an insight that, in this setting, is El Alaoui, Montanari and Sellke's. *same technique*: (5.4) is their planted law conjugated by $D_{x_0} = \operatorname{diag}(x_0)$, the size bias on p. 39 is their Proposition 4.2 with the GOE diagonal split off, and the Markov-inequality transfer is a quantitative form of their contiguity step. The lower tail (5.27), proved anew, follows from the Aizenman–Lebowitz–Ruelle limit [2] quoted on p. 3, credited elsewhere.

Correspondence.

	Preprint	El Alaoui–Montanari–Sellke
Law	(J, X), $X \sim \mu_0$, biased by $L(J) = Z(J)/\mathbb{E}Z(J)$, $Z(J) = Z_0$: X uniform, $J_{ik} \mid X$ independent, mean jX_iX_k/n (p. 39)	μ_{pl}: x uniform, $A = (\beta/n)xx^{\mathsf{T}} + W$, $W \sim \mathrm{GOE}(n)$; A-marginal $\mu_{\mathrm{GOE}}(dA)\, Z_{\mathsf{SK}}(A)$ ((4.1)–(4.5), p. 10)
Path	$Y(t) = tX + B(t)$ (1.4); its law given J is unchanged by the size bias (p. 39)	$\mathbb{Q}$: $A \sim \mathrm{GOE}(n)$, $y \overset{d}{=} tx + \bar{B}$, $x \sim \mu_A$ (4.12); $\mathbb{P}$: $y = tx_0 + B$ (4.9); $d\mathbb{P}/d\mathbb{Q} = Z_{\mathsf{SK}}(A)$ (Prop. 4.2, p. 11)
Planted model	$\tilde{J} = W + (j/n)\mathbf{1}\mathbf{1}^{\mathsf{T}}$, $h_0 = t\mathbf{1} + \sqrt{t}\, g$ (5.4), p. 28	$\mathbb{P}$: $\beta A = (j/n)x_0x_0^{\mathsf{T}} + \beta W$, $y(t) = tx_0 + B(t)$; their proof works under $\mathbb{P}$ (p. 12)
Transfer	$\mathbb{P}\{L(J) < e^{-\delta n}\} \to 0$ (5.27); with Markov, the planted bound $Ce^{-a_4 n}$ gives path failure $\le e^{-an}$ given typical J (pp. 39–40)	$Z_{\mathsf{SK}}(A) \to e^W$ in law, $W \sim N(-\sigma^2, 2\sigma^2)$ (Thm 4.1, p. 11); by Le Cam's first lemma, results under $\mathbb{P}$ transfer to $\mathbb{Q}$ (pp. 11–12)

How to check.

1. The size bias on p. 39 against [1, (4.1)–(4.6), p. 10; Prop. 4.2, p. 11]. For $J = \beta(A - \operatorname{diag} A)$, $\mu_A = \mu_0$ and $Z_{\mathsf{SK}}(A) = 2^{-n}\sum_x e^{\beta\langle x, Ax\rangle/2 - \beta^2 n/4} = L(J)\, e^{\beta \operatorname{Tr} A/2 - j/4}$, as $\mathbb{E}Z(J) = 2^n e^{j(n-1)/4}$; $\beta \operatorname{Tr} A/2 \sim N(0, j/2)$ is independent of J, and tilting by $e^{\beta \operatorname{Tr} A/2 - j/4}$ gives each βA_{ii} the law $N(j/n, 2j/n)$ that the preprint adds by hand.
2. (5.4), p. 28 against [1, (4.3), p. 10; (4.9), p. 11]: $\tilde{J} = D_{x_0}(\beta A)D_{x_0}$ and $h_0 = D_{x_0}y(t)$.
3. The transfer on pp. 39–40 against [1, Thm 4.1, p. 11; p. 12]; (5.27) against p. 3: with $\mathbb{E}Z(J)$ as in item 1, the limit quoted there gives $n^{-1}\log L(J) \to 0$ in probability, hence (5.27).

F2. Stopping the localization before its path estimates can fail, with early stops and terminal failures as one exceptional event

Summary. Huang, Mohanty, Rajaraman and Wu introduce "stopped localization schemes" as a generic tool [3, §6, pp. 25–27], and the preprint does not cite their paper; it credits Chen and Eldan for the variance-retention and Dirichlet-form framework (§1.1, p. 4; §2, p. 7), whose annealing theorem needs the terminal gap almost surely [4, Thm 46, p. 32]. The approach: when the estimate driving variance retention and the terminal inequality hold only on typical paths, stop where the estimate may fail, so that retention holds by construction (their Lemma 6.6, p. 26), and treat early stops and terminal failures as one exceptional event (Lemma 6.8, p. 27); the combination is their insight. The preprint's §2.2 (pp. 7–8) and §2.4 (p. 10) follow it, stopping where the stability of Corollary 5.2 (p. 27) may fail: *adapted*, since the exceptional event is charged by its probability under the path law started from $f^2\mu_0$, not by $\operatorname{osc}(f)^2$ times its probability, which gives a Poincaré inequality instead of a weak one. The moves are older (fragments, not cited for this): Eldan stopped stochastic localization when good events failed [22, §5, p. 26], and El Alaoui, Eldan, Gheissari and Piana split the terminal variance into good and bad paths [23, p. 15]; Davies, Lee, Sandhu and Shi's weak Poincaré paper restates Lemma 6.8 on the hypercube [14, Thm 5.5, p. 22].

Correspondence. Preprint: $\mathbb{E}_{\mu_0} f = 0$, $\mathbb{E}_{\mu_0} f^2 = 1$, $V_t = \mathrm{Var}_{\mu_{Y(t)}} f$, $N_t = \mathbb{E}_{\mu_{Y(t)}} f^2$, $d_0 = \mathcal{D}_0(f)$; $\mathbb{P}^f$, the path law from $X \sim f^2\mu_0$, has density N_τ at τ. Huang, Mohanty, Rajaraman and Wu: μ_t [3, Def. 6.2, p. 25] is $\mu_{Y(t)}$ if $C_t = I$; (ρ, δ)-weak Poincaré: $\mathrm{Var}\,\varphi \le \rho^{-1}\mathcal{E}(\varphi,\varphi) + \delta\,\mathrm{osc}(\varphi)^2$.

	Preprint	Huang–Mohanty–Rajaraman–Wu
Stopping	$\tau = T \wedge \tau_0$, τ_0 the first time a fixed witness (y, A), $\mathsf{H}_J(y, A) \not\succ cI$, has $\|F_{J,Y(t)}(y)\| \le \rho_0\sqrt{n}$ (§2.2, p. 7; proof of Cor. 5.2, p. 27)	$\tau = T \wedge \inf\{t : \|C_t^{1/2}\,\mathrm{Cov}(\mu_t)C_t^{1/2}\|_{\mathrm{op}} \ge K\}$ (Lemma 6.6, p. 26)
Retention	$\mathbb{E}V_\tau \ge v - (1-v)d_0$, $v = e^{-C_{\mathrm{d}}T}$ (2.4), p. 8	$\mathbb{E}\,\mathrm{Var}_{\mu_\tau}[\varphi] \ge e^{-TK}\,\mathrm{Var}_\mu[\varphi]$ (Lemma 6.6)
Bad event	$\{\tau_0 \le T\} \cup \{\tau_0 > T,\ Y(T) \notin \mathcal{T}_n(J)\}$, $\mathcal{T}_n(J) = \{h : \mathrm{Var}_{\mu_h} g \le 2\mathcal{D}_h(g)\ \forall g\}$; $\mathbb{P}(\mathrm{bad}) \le 2e^{-an}$ (2.3), p. 7	covariance bound fails before T (probability $\le \eta_1$) or μ_T has no (ρ,δ)-weak Poincaré inequality ($\le \eta_2$) (Lemma 6.8, p. 27)
Split	good: $\mathbb{E}[\mathbf{1}_{\mathrm{good}}V_\tau] \le 2d_0$ by the terminal gap; bad: $\mathbb{E}[\mathbf{1}_{\mathrm{bad}}V_\tau] \le \mathbb{E}[\mathbf{1}_{\mathrm{bad}}N_\tau] = \mathbb{P}^f(\mathrm{bad}) \le v/4$ (2.9); so $\mathrm{Var}_{\mu_0} f \le 4e^{C_{\mathrm{d}}T}\mathcal{D}_0(f)$ (§2.4, p. 10)	good: terminal inequality, $\mathcal{E}_\mu \ge \mathbb{E}\,\mathcal{E}_{\mu_\tau}$; bad: $\mathrm{Var}_{\mu_\tau}\varphi \le \mathrm{osc}(\varphi)^2$, adding $(\eta_1+\eta_2)\,\mathrm{osc}(\varphi)^2$ (Lemma 4.11, p. 20); so a $(\rho e^{-TK}, e^{TK}(\delta+\eta_1+\eta_2))$-weak Poincaré inequality

How to check.

1. τ and (2.4) in §2.2, pp. 7–8 against [3, Lemma 6.6, p. 26] with $C_t = I$.
2. (2.3), p. 7 and the split in §2.4, p. 10 against the proofs of [3, Lemma 6.8, p. 27] and Lemma 4.11 (p. 20): only the bad-path term differs.
3. §1.1, p. 4 against assumption 2 of [4, Thm 46, p. 32]: a gap almost surely at the stopping time.

F3. Uniform diagonal equivalents for Onsager-shifted GOE inverses

Summary. Jekel, Sandhu and Shi [5, Thm 4.1(3), pp. 30–31] estimate the diagonal of the Onsager-shifted resolvent of a GOE matrix plus a diagonal for all diagonals at once, as their algorithm computes the diagonals from the same matrix; Davies, Lee, Sandhu and Shi [6, Cor. B.35, p. 179] build on this for the squared resolvent. The preprint cites neither, nor Fan, Mei and Montanari's trace form [7, Lemma 4.9, pp. 19–20] of the Onsager-point identity (a fragment). Jekel, Sandhu and Shi's approach, "uniform asymptotic freeness from the diagonal" [5, p. 30], predicts the diagonal by freeness, proves it in expectation by integration by parts and a Poincaré bound, and extends it to all diagonals by concentration over a net, a step they call standard. The preprint's Lemmas 3.1 and 3.3 (pp. 12–19) have the same three-part structure; the relationship is *adapted*, because they predict by Speicher's operator-valued moment rule, which the preprint cites (p. 11), and compute expectations from finite-n loop equations, not by free-model interpolation. Davies et al. (also *adapted*) truncate and chain as the preprint does, and their Lemma B.18(b) is the free-model form of the identity $\mathsf{P}(K_A) = I$ that the preprint proves without citation (p. 16): Speicher's functional equation [8, Thm 4.1.12, pp. 52–53] at an explicit solution.

Correspondence. Here $D = A^{-1}$, so $K_A(J)A = (D + j\chi_A I - J)^{-1}$; the sources' $2\beta A_{\mathrm{sym}}$ and βA have the law of W for $j = 2\beta^2, \beta^2$; $E_{\mathcal{D}_n}$ is the diagonal part, S is free semicircular, $b = \beta n^{-\delta}$.

	Preprint	Jekel–Sandhu–Shi [5]	Davies–Lee–Sandhu–Shi [6]
On one event	Lemma 3.1 (p. 12): for all $0 \le A \le a_+ I$ with $S_A(1; J) \succeq cI$ and words Q, which may contain $K_A(J)A$, $\|\mathrm{diag}(Q - \mathsf{P}(Q))\|_2 \le C$ (Euclidean norm)	Thm 4.1(3) (pp. 30–31): for all $D \ge 0$ with $2\beta^2 \operatorname{tr}_n(D^{-2}) = 1$, $\|E_{\mathcal{D}_n}[P^2] - b(b^2 + D^2)^{-1}\|_2 \le C_3 \beta n^{-2\delta}$, $P^2 = -\operatorname{Im}(\tilde{z} - 2\beta A_{\mathrm{sym}} + D)^{-1}$	Cor. B.35 (p. 179): for all $D \succeq I$, $a \in [0, 1]$, $\|Y - F'(a)D(F(a) + D)^{-2}D\|_{L^2} \le C\beta\gamma^{-4}n^{-1/2}$, $Y = E_{\mathcal{D}_n}[D(a - \beta A + D)^{-2}D]$
Onsager point	$\mathsf{P}(K_A) = I$ (p. 16); (3.18) at $z = 1$: $\mathbb{E}K_{ii} = 1 + O(n^{-1})$	$E_{\mathcal{D}_n}[(\tilde{z} - \sqrt{2}\beta S + D)^{-1}] = (ib + D)^{-1}$, $\tilde{z} = ib + 2\beta^2 \operatorname{tr}_n((ib + D)^{-1})$ (p. 38; Thm 4.7, Prop. 4.11)	$E_{\mathcal{D}_n}[(a - \beta S + D)^{-1}] = D^{-1}$, $a = \beta^2 \operatorname{tr}_n(D^{-1})$ (p. 17 display with Lemma B.18(b), p. 164)
Mean	Integration by parts (3.15), covariance $\le Cn^{-3/2}$ by (3.13), loop equations (3.17)	Free-model interpolation, integration by parts, Poincaré (Prop. 4.20, Lemma 4.24, pp. 43–47)	The same, with a cutoff (Prop. B.23, pp. 169–170)
All diagonals	Projection Π_R onto $\{\|M\| \le R\}$ (p. 13); dyadic meshes (3.21)–(3.22), p. 18	Herbst concentration, ε-net in D^{-1} (Lemmas 4.18–4.19, pp. 40–43)	Projection $\Pi_{C_{\mathrm{op}}}$ onto $\{\|M\|_{\mathrm{op}} \le C_{\mathrm{op}}\}$, chaining (Thm B.9, pp. 160–162)

How to check.

1. [5, Thm 4.1(3), pp. 30–31] and [6, Cor. B.35, p. 179] against Lemma 3.1 (p. 12).
2. [6, p. 17; Lemma B.18(b), p. 164] with $D = A^{-1}$, $\beta^2 = j$, against $\mathsf{P}(K_A) = I$ and (3.18) (p. 16).
3. [5, Prop. 4.20, Lemma 4.24, pp. 43–47] against (3.15)–(3.18) (pp. 15–16).

F4. A planted Kac–Rice first moment for TAP zeros: determinant rate, Gaussian residual density, and the sign of the complexity functional

Summary. Fan, Mei and Montanari bound the expected number of TAP critical points of the planted Sherrington–Kirkpatrick model by a Kac–Rice integral, evaluate its rate as a squared discrepancy minus the relative entropy of the empirical field law to a Gaussian, and show that on the Nishimori line this rate vanishes at the Bayes-consistent law and, for large signal ($\lambda \ge \lambda_0$), is negative near it [7, §3, pp. 8–12; Prop. 5.2, p. 22]. The preprint does not cite this paper and never uses the words Kac–Rice or complexity: its introduction describes the argument only through its own ingredients, "A scalar entropy inequality and a local-volume estimate" (§1.1, p. 4), and calls the integral "a Gaussian integral" (§1.3, p. 5). The computation is the smoothed integral (5.5), p. 28 with Lemma 5.3 (pp. 28–29), the determinant rate of Lemma 3.5 (p. 20), and Lemma 4.1 (p. 22) applied in (5.9), p. 29. Both papers exclude TAP points with the wrong empirical field law by a first moment over the field vector under the planted law, whose integrand is the Gaussian density of the TAP residual times the determinant at its exponential rate and whose rate in $y = \operatorname{arctanh} m$ vanishes at the Nishimori-consistent Gaussian; the insight that this planted functional localizes the TAP points is Fan, Mei and Montanari's. *adapted*: at $t = \sigma = 0$ the preprint's rate functional is

theirs exactly, but the preprint proves a global inequality with its equality case where they prove local negativity at large signal, smooths the residual, and uses the determinant rate as a weight backed by a high-probability bound. Celentano, Fan and Mei [11, Lemma 4.7, p. 17; App. B.2, pp. 41–43] use one step of the approach, and Belius, Concetti and Genovese [24, Thms 1.1–1.2, pp. 2–3] and Boursier [25, Lemma 2.7, p. 20] reprove the determinant rate; none of the three is cited.

Setup. Here $j = \beta^2 < 1$, and F_{J,h_0} and V_y are as in Section 2. The planted law is $\tilde{J} = W + \frac{j}{n}\mathbf{1}\mathbf{1}^\mathsf{T}$, with W a centered GOE of scale j, and $h_0 = t\mathbf{1} + \sqrt{t}\, g$ (5.4), p. 28. For $y \in \mathbb{R}^n$ put $m = \tanh y$, $q = n^{-1}\sum_i m_i^2$, $b = 1 - q$, $B = jb$, $M = n^{-1}\sum_i m_i$ and, for $\sigma > 0$,

$$d = t + jM, \quad s = t + \sigma^2 + jq, \quad S = s + jq, \quad a_y = \frac{1}{n}\sum_i (y_i - d)m_i, \quad \psi = \frac{j(a_y - sb)^2}{2sS}$$

(5.6), p. 28; $N(d, s)$ has variance s, $D(\cdot\|\cdot)$ is relative entropy, and P is the empirical law of the y_i (or any law with finite second moment). The integral of (5.5) is

$$I_{t,\sigma}(\mathcal{E}) = (2\pi\sigma^2)^{-n/2} \int \mathbf{1}_{\{(y,\tilde{J})\in\mathcal{E}\}}\, e^{-\|F_{\tilde{J},h_0}(y)\|^2/(2\sigma^2) + njb(y)^2/2}\, dy,$$

and $F' = F'_{\tilde{J},h_0}(y) = I + (BI - \tilde{J})V_y - \frac{2j}{n} m m^\mathsf{T} V_y$ (Lemma 5.6, p. 35; cf. p. 38). Fan, Mei and Montanari observe $Y = \frac{\lambda}{n} x x^\mathsf{T} + W'$ with $W' \sim \mathrm{GOE}(n)$ (off-diagonal variance $1/n$) and take $x = \mathbf{1}$; $g_n(m)$ and $H_n(m)$ are the scaled gradient and Hessian of the TAP free energy at inverse temperature β ((3.2)–(3.3), p. 8), $Q(m) = \|m\|^2/n$, $M(m) = \langle \mathbf{1}, m\rangle/n$, $A(m) = n^{-1}\sum_i m_i \operatorname{arctanh} m_i$, $E(m)$ is a fourth statistic, ρ is the empirical law of the m_i, and $H(\cdot\,|\,\cdot)$ is relative entropy. With $\beta = \lambda = \sqrt{j}$ (their Nishimori line), $\sqrt{j}\, W' = W$, $y = \operatorname{arctanh} m$ and $t = 0$, one has $g_n(m) = F_{\tilde{J},0}(y)$ and $F' = H_n(m)V_y$.

Role in the argument. The integral bounds the expected weight of unstable configurations, and the determinant rate turns an unstable approximate zero into a lower bound for it: Lemma 4.1 → (5.9) → Lemma 5.4 → (5.25), p. 37; with Lemma 3.5 and Lemma 5.6, Markov's inequality excludes every witness (5.26) → Proposition 5.1 → Corollary 5.2 → the preprint's §2 → Theorem 1.1. The neighbouring steps are findings F7 (the conditional matrix factor in Lemma 5.3), F5 (local volume), F13 (the equality law, Lemma 4.2) and F6 (the route of Proposition 5.1).

The earlier technique. Fan, Mei and Montanari's Theorem 1.1 bounds the growth rate of $\mathbb{E}[\mathrm{Crit}_n(U, V_n)]$, the expected number of TAP critical points with $(Q, M, A, E) \in U \subseteq [\eta, 1] \times \mathbb{R}^3$ and $m \in V_n^n$ (V_n an interval exhausting $(-1, 1)$), by $\sup_U S_\star$ [7, Thm 1.1, p. 4]; they present the proof as the Kac–Rice complexity strategy of earlier work on spherical models (p. 4). In their §3, the Kac–Rice bound $\mathbb{E}[\mathrm{Crit}_n(U, V)] \le \int_{V^n} \mathbf{1}_U\, \mathbb{E}[|\det H_n(m)| \mid g_n(m) = 0]\, p_m(0)\, dm$ (Prop. 3.1, p. 9), the determinant rate $L(m) = \frac{\beta^2(1-Q(m))^2}{2} + \frac{1}{n}\sum_i \log\frac{1}{1-m_i^2}$ (Prop. 3.2, p. 9, proved in their §4 through the free convolution) and the gradient density, a Gaussian mixture over one auxiliary scalar (their y) (Lemma 3.3, p. 9), give the integral $T(U, V_n)$ of Cor. 3.4 (pp. 9–10). Sanov's theorem and a Varadhan bound (Prop. 3.5, p. 10), then the supremum over their y, give the rate $\frac{1}{4\beta^2}(\beta^2(1-q) + \beta\lambda\varphi^2/q - a/q)^2 - H(\rho\,|\,\rho_{\varphi,q})$ (§3.4, p. 11), with $\rho_{\varphi,q}$ the law of $\tanh z$, $z \sim N(\beta\lambda\varphi, \beta^2 q)$; Lagrange multipliers for the constraints give $S_\star$ (p. 12). On the Nishimori line $\beta = \lambda$, Proposition 5.2 (p. 22) shows $S_\star(q_\star, \varphi_\star, a_\star, e_\star) = 0$ for every $\lambda > 0$ and, for $\lambda \ge \lambda_0$ and $|q - q_\star|, |\varphi - \varphi_\star| \le c_0/\lambda^2$, $S_\star \le -c(\lambda)\|(q, \varphi, a, e) - (q_\star, \varphi_\star, a_\star, e_\star)\|^2$; here $q_\star = 0$ when $\lambda \le 1$ (Prop. 5.5, p. 23).

Correspondence.

	Preprint	Fan–Mei–Montanari [7]
Determinant rate	$\frac{1}{n}\log\det(F'^{\mathsf{T}}F'+\gamma I)\le jb^2+\varepsilon$ with high probability, simultaneously over y (Lemma 3.5, (3.26))	$\mathbb{E}[\lvert\det H_n(m)\rvert \mid g_n(m)=0]\le e^{nL(m)+o(n)}$ for $Q(m)\ge\eta$, $m\in V_n^n$ (Prop. 3.2, (3.5), p. 9)
First moment	$\mathbb{E}I_{t,\sigma}(\mathcal{E})$, weight $e^{njb^2/2}$ (5.5)	$T(U,V_n)=$ $e^{o(n)}\int_{(-1,1)^n}\mathbf{1}_U\,e^{nL(m)}p_m(0)\,dm$ (Cor. 3.4, pp. 9–10)
Residual density	$\mathrm{Cov}(Wm)=jqI+\frac{j}{n}mm^{\mathsf{T}}$; $\mathbb{E}I_{t,\sigma}=\int\sqrt{s/S}\prod_i\varphi_{d,s}(y_i)\,e^{n\psi}\cdots dy$ (Lemma 5.3)	$\mathbb{E}[(W'm)_i(W'm)_k]=Q\delta_{ik}+\frac{m_im_k}{n}$ (Lemma 3.3, p. 9); supremum over their y (§3.4, p. 11)
Rate	$\psi-D(P\Vert N(d,s))$ (5.6), (5.9)	squared term $-H(\rho\,\vert\,\rho_{\varphi,q})$ (§3.4, p. 11)
Sign	$D(P\Vert N(d,s))\ge\psi$ for every P with finite second moment when $s\ge jq$; equality iff $P=N(d,s)$ (Lemma 4.1, (4.1))	$S_\star=0$ at the Bayes point; local negativity for $\lambda\ge\lambda_0$ (Prop. 5.2, (5.4), p. 22)

At $t=\sigma=0$, $\beta=\lambda=\sqrt{j}$ and $\varphi=M$, one has $d=jM=\beta\lambda\varphi$, $s=jq=\beta^2q$, $S=2jq$ and $a_y=a-jM^2$ with $a=n^{-1}\sum_i y_im_i=A(m)$, so

$$\psi=\frac{(a-jM^2-jq(1-q))^2}{4jq^2}=\frac{1}{4\beta^2}\Big(\beta^2(1-q)+\frac{\beta\lambda\varphi^2}{q}-\frac{a}{q}\Big)^2,\qquad D(P\Vert N(jM,jq))=H(\rho\,\vert\,\rho_{\varphi,q}),$$

the second because tanh is a bijection carrying $N(\beta\lambda\varphi,\beta^2q)$ to $\rho_{\varphi,q}$. For the determinant, $F'=H_n(m)V_y$ and $dm=\prod_i(1-m_i^2)\,dy$ give $e^{nL(m)}\,dm=e^{njb^2/2}\,dy$: the rate $jb^2/2$ of $\frac{1}{n}\log\lvert\det F'\rvert$ in Lemma 3.5 is $L(m)$ without its Jacobian term $\frac{1}{n}\sum_i\log\frac{1}{1-m_i^2}$, and $T(U,V_n)$ is, up to $e^{o(n)}$, the integral $\int_{\mathbb{R}^n}\mathbf{1}_U\,e^{njb^2/2}p_{\tanh y}(0)\,dy$, the $\sigma\to0$ limit of $\mathbb{E}I_{0,\sigma}$ restricted to $(Q,M,A,E)\in U$.

Differences.

- The sign is proved for every P with finite second moment whenever $s\ge jq$, with its equality case, by the transport $T_u(y)=y-u\tanh y$ (4.4), p. 23; Fan, Mei and Montanari's negativity holds only within c_0/λ^2 of the Bayes point, for $\lambda\ge\lambda_0$ and $Q\ge\eta>0$.
- The regime is $j<1$ (their $\beta=\lambda<1$, where $q_\star=0$), with $t\ge0$ and $\sigma>0$; the Gaussian reduction is an exact identity with prefactor $\sqrt{s/S}$, not a supremum over their y.
- Lemma 3.5 is an unconditional high-probability bound on $\log\det(F'^{\mathsf{T}}F'+\gamma I)$ for the non-symmetric F', simultaneous over diagonal V and finite-rank perturbations and proved by interpolation in z (3.27)–(3.29); Fan, Mei and Montanari bound $\mathbb{E}[\lvert\det H_n(m)\rvert \mid g_n(m)=0]$ at each fixed m.

How to check.

1. (5.6) and (5.9) at $t=\sigma=0$ against [7, §3.4, p. 11], via the display after the table.
2. The proof of Lemma 5.3 against [7, Lemma 3.3 and Cor. 3.4, pp. 9–10] after the substitution $m=\tanh y$.

3. Lemma 3.5, with F' as on p. 38, against [7, Prop. 3.2, (3.5), p. 9] using $F' = H_n(m)V_y$.
4. Lemma 4.1 (p. 22) against [7, Prop. 5.2, p. 22].

F5. Excluding approximate zeros by a smoothed first moment and local volume

Summary. Huang and Sellke [9, §5, Thm 5.2, Lemma 5.11, pp. 28–36] (2023) show how an annealed bound on exact critical points with a bad property rules out, with probability $1 - e^{-cN}$, approximate ones with that property; the preprint cites them nowhere, calls the step a "local-volume estimate" (§1.1, p. 4) and a "local-volume lower bound" (§1.3, p. 5) without citation, and never mentions Kac–Rice. Their insight: approximate zeros cannot be counted, so one integrates a narrow Gaussian in the residual, weighted by the Jacobian determinant (in Huang and Sellke, its conditional expectation, of known exponential rate), over the bad set and bounds its mean; near an approximate zero the Gaussian volume from the linearization is $|\det|^{-1}$, cancelling the weight wherever the determinant is at most its typical rate (everywhere at once, with high probability), so the integral is at least $e^{-o(n)}$, and Markov's inequality concludes. The preprint follows this in (5.5), Lemma 5.6 and the proof of Proposition 5.1, with Lemma 3.5 as the uniform determinant bound; *adapted*, because only the external field is rerandomized, the deterministic weight $e^{njb(y)^2/2}$ replaces the determinant, and regularizing the Jacobian L to $L^\mathsf{T}L + \gamma I$ replaces their spectral split. Huang, Montanari and Pham [10, p. 29], also not cited, use a consequence of Theorem 5.2 (finding F6).

Correspondence.

	Preprint	Huang–Sellke
Smoothed integral	(5.5), p. 28: $I_{t,\sigma}(\mathcal{E}) = (2\pi\sigma^2)^{-n/2}\int_{\mathcal{E}} e^{-\|F(y)\|^2/(2\sigma^2)+njb(y)^2/2}\,dy$, $F = F_{\tilde{J},h_0}$; by the area formula, the expected number of zeros of $F_{\tilde{J},h_0+\sigma\xi}$ in $\mathcal{E}$, with $e^{njb(y)^2/2}$ for $\lvert\det L\rvert$	given H_N, the density at 0 of $\nabla_{\mathrm{sp}}H_{N,\delta}(y)$, $H_{N,\delta} = \sqrt{1-\delta}\,H_N + \sqrt{\delta}\,H'_N$, times $\mathbb{E}[\lvert\det\nabla^2_{\mathrm{sp}}H_{N,\delta}(y)\rvert \mid \cdots] = e^{N\int\log\lvert\lambda\rvert\,d\mu\pm o(N)}$ ((5.20)–(5.22), pp. 33–34)
Local lower bound	Lemma 5.6: $\|F(y)\| \le \rho\sqrt{n}$ and $\frac{1}{n}\log\det Q \le jb(y)^2 + \varepsilon_1$, $Q = L^\mathsf{T}L + \gamma I$, give $\ge e^{-\varepsilon n}$ over a ball (5.19)–(5.20), p. 35; the ellipsoid $y + \sigma Q^{-1/2}g''$ gives $(\det Q)^{-1/2}$ (5.21), p. 36, cancelling $e^{njb(y)^2/2}$ (p. 37)	Lemma 5.11 (p. 33): from $\|\nabla_{\mathrm{sp}}H_N(x)\| \le \varepsilon\sqrt{N}$ (5.15), $\ge e^{-o_\delta(N)}$ expected critical points of $H_{N,\delta}$ in $B_{\alpha\sqrt{N}}(x)$; the volume $\mathrm{Vol}(S_1)\lvert\det(J_N(x)\rvert_{S_2})\rvert^{-1}$ ((5.23), p. 35) cancels (5.21)
Uniform determinant	Lemma 3.5: with probability $\ge 1 - Ce^{-c_4 n}$, $\frac{1}{n}\log\det(L^\mathsf{T}L + \gamma I) \le jb^2 + \varepsilon$ for all diagonal $0 \le V \le I$ and bounded rank-r_0 E at once (3.26), p. 20; used at every y (p. 38)	$H_N \in K_N(\varepsilon)$ (Def. 13(c), Prop. 5.8, p. 32): with probability $\ge 1 - e^{-c(\varepsilon)N}$, for all x the spectrum of $J_N(x)$ is W_1-close to $\tilde{\mu}_{\xi,\nabla_{\mathrm{rad}}H_N(x)}$
Conclusion	a witness (5.26), p. 37 forces $I_{t,\sigma}(E_t) \ge e^{-\varepsilon n}$ for the unstable set E_t; Markov and $\mathbb{E}I_{t,\sigma}(E_t) \le e^{-a_0 n}$ (5.25), p. 37 bound its probability by $Ce^{-a_2 n}$ (p. 38)	(5.24), averaged, and $\mathbb{E}\lvert\mathsf{Crt}_N^{\overline{\mathcal{D}},0,\upsilon/2}(H_N)\rvert \le e^{-c_0 N}$ (5.3) bound the probability of a bad approximate critical point by e^{-cN} (p. 36)

How to check.

1. (5.5) and Lemma 5.6, pp. 28 and 35–37 against [9, (5.20)–(5.23), Lemma 5.11, pp. 33–35].
2. Lemma 3.5 and pp. 37–38 against [9, Def. 13(c), Prop. 5.8, p. 32; (5.24), p. 36].

F6. TAP stability along a localization path by a planted first moment

Summary. Huang, Montanari and Pham prove, for spherical spin glasses at a fixed time of the stochastic-localization path, that with probability $1 - e^{-cN}$ the TAP free energy has a unique well-conditioned critical point in a band, near which every approximate critical point lies [10, Prop. 6.1, pp. 26–27]; the preprint does not cite that paper. Their route gets well-posedness from a first moment under the planted law of El Alaoui, Montanari and Sellke: bound the bad critical points by a conditional probability of a bad Hessian times a subexponential total, pass to approximate critical points, and transfer back through the normalized partition function. The preprint follows this route for the field equation at the fields its observation path visits, in Proposition 5.1 (p. 27) with Lemma 5.7 (p. 40); its introduction attributes this uniformity to its scalar entropy inequality and local-volume estimate (§1.1, p. 4) without naming an earlier instance of the route. The relationship is *adapted*: the total comes from the entropy inequality, not topological trivialization; the bound is uniform in $t \le T$ through a fixed mesh; uniqueness of the root is global and deterministic; the transfer is quenched, not by contiguity; and approximate roots are handled neither by the theorem of Huang and Sellke that Huang, Montanari and Pham cite nor by the device behind it, which rerandomizes the disorder [9, (5.1) and Thm 5.2, p. 28], but by the preprint's own device on the same principle (an approximate zero forces annealed mass $e^{-o(n)}$): a residual-smoothed integral, in effect a perturbation of the field, with a local-volume lower bound (finding F5).

Correspondence.

Step	Preprint	Huang–Montanari–Pham
Conclusion	with $F = F_{J,Y(t)}$, (5.3), p. 27: for $J \in \mathcal{J}_n$, conditional probability $\ge 1 - e^{-an}$, all $t \le T$, $\|F(y)\| \le 2\rho_0\sqrt{n} \Rightarrow \mathsf{H}_J(y, A) \succ cI$ for diagonal A near V_y; (5.29), p. 40: one zero r, and $\|y - r\| \le C\|F(y)\|$ if $\|F(y)\| < 2\rho_0\sqrt{n}$	Prop. 6.1, pp. 26–27: under $\mathbb{P}$, one t, probability $1 - e^{-cN}$, a unique critical point m^{TAP} in the band $\mathcal{S}_\iota$, Hessian spectrum in $[-C_{\max}, -C_{\min}]$; every ι-approximate critical point in $\mathcal{S}_\iota$ is within $o_\iota(1)$ of it in $\|\cdot\|_N = \|\cdot\|/\sqrt{N}$
Planted first moment	(5.25), p. 37, under (5.4), p. 28: $\mathbb{E} I_{t,\sigma}(E_t) \le e^{-a_0 n}$ for the smoothed integral over unstable y, a conditional probability $e^{-a_* n}$ (Lemma 5.5) times a scalar integral $e^{o(n)}$ (Lemma 5.4(i))	under $\mathbb{P}$ ((3.2), p. 12), Kac–Rice (6.6) and Cauchy–Schwarz: $\mathbb{E}\lvert\mathrm{Crt} \cap \mathcal{T}\rvert \le C \sup \mathbb{P}[m \in \mathcal{T} \mid \cdots]^{1/2}$, using $\mathbb{E}\lvert\mathrm{Crt}\rvert = 2 + o_N(1)$ (Lemma 6.8, Prop. 6.7, p. 30; Prop. 6.9, p. 31)
Approximate roots	a witness (5.26) forces $I_{t,\sigma}(E_t) \ge e^{-\varepsilon n}$ (Lemma 5.6); Markov's inequality (p. 38)	Huang and Sellke's Theorem 1.6 [9, p. 8], in the proof of Prop. 6.2(a) (p. 29)
Transfer	$L(J) = Z(J)/\mathbb{E}Z(J)$, lower tail (5.27), p. 39, Markov's inequality (pp. 39–40)	$d\mathbb{P}/d\mathbb{Q} = Z(H_N)/\mathbb{E}Z(H_N)$, contiguity (Prop. 3.4, Cor. 3.5, p. 12; used p. 93)

How to check.

1. (5.3), p. 27 and (5.29), p. 40 against [10, Prop. 6.1, pp. 26–27; (4.8)–(4.9), p. 15].
2. The plan on p. 27 and §5.4 against §3.2 and the proofs of Props. 6.1–6.2 [10, pp. 11–12, 27–32]; for approximate roots (pp. 37–38), also against [9, (5.1), Thm 5.2, p. 28].

F7. Conditioning the TAP Hessian on the residual near the Bayes law to exclude unstable zeros

Summary. Celentano, Fan and Mei prove local strong convexity of the TAP free energy for $\mathbb{Z}_2$ synchronization by "a second Kac–Rice upper bound" [11, §4.1, p. 14]: the event that the Hessian's least eigenvalue near m is small enters the count of critical points as a probability conditional on a vanishing gradient (Lemma 4.8); conditioning leaves a fresh projected GOE plus explicit low-rank terms (Lemma B.2); and near the Bayes (Nishimori) law this probability is exponentially small, by a finite-dimensional Schur complement whose positivity rests on $1-\lambda^2\mathbb{E}[(1-m^2)^2] > 0$ (Lemma 4.9, Cor. 4.10, p. 18). The preprint does not cite this paper, and cites no other work for this step. The marked Kac–Rice bound itself is generic: they derive it from Fan, Mei and Montanari's arguments and a theorem of Adler and Taylor [11, App. A.5, pp. 33–34]. What is theirs is its combination with the TAP equation and the Bayes law. The preprint follows that combination: in an exact Gaussian identity for a smoothed count of approximate zeros, the instability event enters as a probability conditional on the residual (5.8), p. 28, and Lemma 5.5 (pp. 31–35) bounds it by $e^{-a_* n}$ near the equality laws $N(s_*, s_*)$ (finding F13), giving (5.25), p. 37. *adapted*: the conditioning is on a noisy residual, their Sudakov–Fernique variational bound becomes whitening plus an exact Schur complement whose determinant equals theirs up to a positive factor, and positivity is simultaneous over nearby diagonal weights; Huang, Montanari and Pham later take the same step for spherical TAP along stochastic localization [10, Lemma 6.8, Prop. 6.9, pp. 30–31].

Correspondence.

	Preprint	Celentano–Fan–Mei
First moment	$\mathbb{E}I_{t,\sigma}(\mathcal{E}) = \int \sqrt{s/S}\, G_{t,\sigma,n}(y)\, \mathbb{P}\big((y,\tilde{J}) \in \mathcal{E} \mid Wm + \sqrt{t+\sigma^2}\, g' = z\big)\, dy$ (5.8), p. 28	$\mathbb{E}\lvert\mathcal{C}\rvert \le \int_T \mathbb{E}\big[\lvert\det H(m)\rvert\, \mathbf{1}\{\ell(m,W)+U < t\} \;\big|\; g(m)=0\big]\, p_{g(m)}(0)\, dm$ ((4.18), p. 18)
Conditional law	$W = P_u W_0 P_u + uw^\mathsf{T} + wu^\mathsf{T} + \alpha uu^\mathsf{T}$, $u = m/\lVert m\rVert$, $w = \bar{w} + \kappa P_u W_0 u$, W_0 a fresh GOE (5.12), p. 32	given $g(m)=0$, W is $P_m^\perp \tilde{W} P_m^\perp$, $\tilde{W}$ a fresh GOE, plus rank-two terms in m, $z(m)$ (Lemma B.2, pp. 43–44)
Bayes law	(P_y, t) near $\{(P_t, t)\}$, P_y the empirical law of y, $P_t = N(s_*, s_*)$ (Lemma 5.5, p. 31)	$\mathcal{D}_\eta$: $W(\hat{\mu}_m, \mu_\star) < \eta$, with $\operatorname{arctanh} m \sim N(\lambda^2 q_\star, \lambda^2 q_\star)$ under $\mu_\star$ ((4.5)–(4.8), p. 14)
Sign	Gram matrix H (5.16), p. 33; $\det(I-HC) = (d_0+2g_1)(d_0+g_1)^2 > 0$ (5.17), p. 34	Schur complement along m, $\mathbf{1}$: $c_2 > 0$, $c_1 - c_2 > 0$ ((B.13)–(B.14), pp. 39–40, 48–49)

How to check.

1. (5.8), p. 28 and the end of its proof (p. 29) against [11, Lemma 4.8, pp. 17–18].
2. (5.12), p. 32 at $t = \sigma = 0$ against [11, Lemma B.2, pp. 43–44], with $\lambda^2 = j$, $\lambda W \leftrightarrow W$, $z = z(m)$.
3. (5.16)–(5.17), pp. 33–34 against [11, (B.13)–(B.14), pp. 39–40]: with $\lambda^2 = j$, $c_2(c_1 - c_2) = (1-jf)^2(1-jf+je)/(fe) = \det(I-HC)/(fe)$, where $f = \mathbb{E}[(1-m^2)^2]$ and $e = \mathbb{E}[m^2(1-m^2)]$ under the Bayes law.

F8. A memory correction for every input of a multi-input recursion, and its cancellation of one-step reversals at every depth

Summary. Bayati, Lelarge and Montanari (BLM) correct each input of a multi-input nonlinearity in approximate message passing (AMP) by its averaged partial derivative times that input's own source [12, (1.4)–(1.6), p. 7], and prove that these memory terms cancel, at every depth, the one-step reversing contributions to the expansion of the iterates over labelled trees (§2, §4); Montanari [13, (2.1), p. 6] writes the rule for every earlier iterate. The preprint cites neither paper: it cites Thouless, Anderson and Palmer for the Onsager correction (§1.1, p. 4), credits (6.1) to Bolthausen (§6.1, p. 42), whose recursion has a single memory term with a deterministic coefficient [26, p. 4], and names AMP only in §1.1, citing El Alaoui, Montanari and Sellke and Celentano for other content. Its rule (6.7) in Definition 6.2 (p. 44) is Montanari's (2.1) term for term after rescaling; it is used again for the implicit field (6.14), p. 46, and in the proof of Lemma 6.5, (6.33), p. 54, and the cancellation runs through the proofs of Lemmas 6.1 (p. 43), 6.4 (pp. 51–52) and 3.3 (p. 16). The shared insight, made rigorous by BLM: in a recursion in a symmetric random matrix, the averaged derivative times an input's source is minus the leading contribution of a step that immediately reverses, so subtracting it for each input frees each coordinate from its own back-reaction. The relationship is *adapted*: the preprint cancels the diagonal noncrossing-pairing predictions of formal spin-flip Jacobians, not moments of the iterates, by a signed inclusion–exclusion over adjacent pairs, not by comparison with non-backtracking iterates; Montanari's paper holds the rule alone.

Correspondence.

	Preprint	BLM; Montanari
Rule	$y^a = Jw^a - j\sum_l \langle \partial_l w^a\rangle m^{l-1}$ $-j\sum_{b<a}\langle \partial_{y^b} w^a\rangle w^b$ (6.7), p. 44, $\langle\cdot\rangle$ the coordinate average	$u^{k+1} = Af_k(u^0,\dots,u^k) - \sum_{r=1}^k b_{k,r} f_{r-1}$, $b_{k,r} = n^{-1}\sum_i \partial f_k/\partial u_i^r$ (Montanari (2.1)); $-\sum_k A_{ik}^2\, \partial_x f^k(x_k^t)\, f^i(x_i^{t-1})$ (BLM (1.6))
Linear	formal Jacobians (6.3) with every diagonal I: $H_{l+1} = JH_l - jH_{l-1}$, $H_2 = J^2 - jI$ (p. 42)	$f(x) = x$: $x^{t+1} = Ax^t - x^{t-1}$, $x^2 = A^2\mathbf{1} - \mathbf{1}$, cancelling the paths $1 \to i \to 1$ (p. 11)
One step	$-jb_sI$ is minus the contraction of $JD_{1-(m^s)^2}J$ (p. 43); an example on p. 51	$f(x) = x^3$: b_1 cancels the trees with a label equal to the root's ((2.6), pp. 11–12)
Every depth	each subset T of the chain's adjacent pairs contracted, sign $(-1)^{\lvert T\rvert}$: per full noncrossing pairing, $(1-1)^a = 0$ (pp. 43, 52)	$x_i^t = z_i^t$ (non-backtracking trees, Lemma 1, p. 19) plus trees with a backtracking 3-path or star (Lemma 3, p. 26; (4.29), p. 27), negligible by (4.30), p. 28

How to check.

1. (6.7), p. 44 against [13, (2.1), p. 6] with $J = \sqrt{j}\,A$, fields divided by $\sqrt{j}$; and [12, (1.5)–(1.6), p. 7].
2. The proof of Lemma 6.1 (p. 43) with all diagonals I against [12, (2.3), pp. 10–11].
3. "Cancellation of the formal traces" (pp. 51–52) against [12, Lemma 3, (4.29)–(4.30), pp. 26–29].

F9. A gap at a large observation time from few weak-field coordinates and uniformly small principal submatrices

Summary. Davies, Lee, Sandhu and Shi's Potential Hessian Ascent III (PHA III, May 2026) studies the SK posterior $\mu_{\beta A,y_T}$ at a large constant localization time, the preprint's $\mu_{Y(T)}$ ($A \sim \mathrm{GOE}(n)$, $J = \beta(A - \operatorname{diag} A)$ in law), and states the insight [6, p. 19]: only $O(\varepsilon n)$ coordinates can still fluctuate, for any prior (Lemma 7.43), so only small principal submatrices of the disorder matter, and a Gaussian norm tail with a union bound over supports makes them all small at once (Lemma 7.23). The preprint cites neither PHA III nor Davies, Lee, Sandhu and Shi's weak Poincaré paper, whose Theorem 5.1 [14, p. 21] uses the same two lemmas for a weak Poincaré inequality for Glauber dynamics at $\mu_{\beta A,y_T}$, $\beta < 1/2$; at the lead-in to Lemma 8.2 (p. 61) it cites only Bandeira, El Alaoui and Rödder's later Theorem 1.1 [15], for combining small-submatrix bounds with a large constant field. Lemma 8.2 and the proof of Proposition 8.3 (pp. 63–64) carry out the mechanism at $Y(T)$ and obtain $\mathrm{Var}_{\mu_{Y(T)}} f \le 2\mathcal{D}_{Y(T)}(f)$ for any law of the spins (*adapted*: both probabilistic steps also appear in PHA III, but its functional-inequality step on a Hamming ball, through the needle decomposition of $\beta A + \gamma I$ that forces $\beta < 1/2$, is replaced by a split of the signed curvature form over a configuration-dependent weak set).

Correspondence.

	Preprint	PHA III; weak Poincaré paper
Why small submatrices	the few coordinates with appreciable conditional variance occupy a small principal submatrix (p. 61)	on a wedge only $O(\varepsilon n)$ coordinates can fluctuate, so small principal submatrices matter, not A (p. 19)
Few weak coordinates, any prior	X of any law, $Y = Y(T)$: given X, the events $\{\lvert Y_i\rvert < T/2\}$ are independent with probability $p_T \le e^{-T/8}$; exponential Markov: at most $\alpha n/2$ occur, off probability $e^{-\alpha n/4}$ (p. 63)	$x_0 \sim \mu$ of any law, $y_t = tx_0 + B_t$: the events $\{(x_0)_i \ne \operatorname{sign}(y_t)_i\}$ are independent with probability $\Phi(-\sqrt{t})$; Chernoff (Lemma 7.43, p. 135)
All small submatrices at once	$\lVert U_{SS}\rVert, \lVert Q_{SS}\rVert \le \varepsilon$ for all $\lvert S\rvert \le \alpha n$, where $U = \tanh J$, $Q = U \circ U$ (8.11), p. 62; for U, a net on each support and a union bound over the supports of size $\lfloor \alpha n \rfloor$ (pp. 62–63)	$\mathbb{P}(\forall \lvert S\rvert \le \varepsilon n : \lVert A_{S\times S}\rVert \le c_1\sqrt{h(\varepsilon)}) \ge 1 - 2e^{-h(\varepsilon)n}$, h the binary entropy, by the GOE norm tail and a union bound over supports (Lemma 7.23, p. 115)
At the terminal measure	$S(x) = \{i : \lvert Y_i + (Jx)_i\rvert < T/4\}$ has $\lvert S(x)\rvert \le \alpha n$ for every x (8.16), p. 64; split the curvature form over $S(x)$ (p. 64)	$\beta < 1/2$: entropy contraction $\ge \rho_\beta/n$ for the polarized walk on Hamming balls of radius $\varepsilon(\beta)n$ (Thm 7.1, p. 110); additive error e^{-Cn} (Thm 5.1, p. 21)

How to check.

1. The count (p. 63) against the proof of [6, Lemma 7.43, p. 135].
2. The U_{SS} part of (8.11) and its proof (pp. 62–63) against [6, Lemma 7.23 and proof, p. 115].
3. The lead-in (p. 61) and the split (p. 64) against PHA III's overview (p. 19), Theorem 7.1 and Remark 7.2 (p. 110); (8.14) against [14, Thm 5.1, p. 21].

F10. A Poincaré inequality from covariance bounds at the localization fields, obtained from a uniformly positive TAP Hessian

Summary. El Alaoui and Gaitonde [16, p. 2 and §2, pp. 3–4] proposed a route to Poincaré and log-Sobolev inequalities at every $\beta < 1$: bound the covariance of the tilted measure μ_y in operator norm at the fields of Eldan's stochastic localization, through its approximation by the inverse negative TAP Hessian and a negative bound on that Hessian at the mean, then transfer the bound along the localization following Chen–Eldan. The preprint cites them only for their zero-field Theorem 1.1 (§1.1, p. 4), credits Chen–Eldan and Celentano's local convexity but no work for the route joining them, and presents its posterior estimates as "additional" to covariance results, which "control linear fluctuations" (p. 4). It carries out the route: Proposition 5.1 (p. 27) bounds H_J below, and $\mathsf{H}_J(y, V_y)$ is their negative TAP Hessian in normalized coordinates; (1.5), proved as Proposition 7.2 (p. 56), is equivalent up to constants to $\|\mathrm{Cov}_{\mu_h}(x)\| \le C'$ at every good field, which the preprint never states; and its §2 (pp. 7–10) is the transfer, adding the square-reweighted mean (1.6) and a terminal gap. The relationship is *same route*: they state each step heuristically, and the preprint proves each by its own means, the covariance bound by residual identities tested against an implicit linear-response vector rather than through their Frobenius bound (2.7); Brennecke, Xu and Yau [17, Thm 1.4, p. 6], also not cited, had carried out that step at homogeneous fields $h\mathbf{1}$.

Correspondence. Their field y is the preprint's h (h_0 in its §§5–7); in the preprint column, y is the variable of (5.1)–(5.2) and $m = \tanh y$.

Step	Preprint	El Alaoui–Gaitonde [16]
Hessian	$\mathsf{H}_J(y, V_y) = V_y^{1/2}\Phi''(m)V_y^{1/2}$ (proof of Lemma 5.7, p. 40)	$-\nabla^2 F_{\mathrm{TAP}}(m)$ at a maximizer m, (2.2)–(2.5), p. 3
Margin	$\mathsf{H}_J(y, A) \succ cI$ for $\|F_{J,Y(t)}(y)\| \le 2\rho_0\sqrt{n}$, A near V_y, $t \le T$, off probability e^{-an} (Prop. 5.1, p. 27)	$\lambda_{\max}(\nabla^2 F_{\mathrm{TAP}}(m)) \le -\varepsilon$ at the mean m of μ_y, with probability $1 - O(1/n)$ (p. 4)
Covariance	(1.5), i.e. (7.2), at every field satisfying (5.3) (Prop. 7.2, p. 56); equivalently $\|\mathrm{Cov}_{\mu_h}(x)\| \le C'$	Bounded $\mathbb{E}\|\mathrm{Cov}(\mu_y)\|_{\mathrm{op}}$, via the Frobenius form (2.7) of $\mathrm{Cov}(\mu_y) \approx (-\nabla^2 F_{\mathrm{TAP}}(m))^{-1}$ (p. 4)
Transfer	$u'(t) = -\mathbb{E}[\mathbf{1}_{\{t<\tau\}}\|\mathrm{Cov}_t(f, x)\|^2]$, bounded through (1.5) (§2.2, p. 8)	Bounds on $\|\mathrm{Cov}(\mu_y)\|_{\mathrm{op}}$ at localization fields give Poincaré and log-Sobolev for μ (p. 4)

How to check.

1. [16, (2.3), (2.5), p. 3] against (5.2), p. 27 and Φ in the proof of Lemma 5.7, p. 40: with $J = \beta A$, $j = \beta^2$ and $D(m) = V_y^{-1}$, $\Phi(m) = -F_{\mathrm{TAP}}(m) - h_0^{\mathsf{T}} m + n(\log 2 + j/4)$ and $\mathsf{H}_J(y, V_y) = V_y^{1/2}(-\nabla^2 F_{\mathrm{TAP}}(m))V_y^{1/2}$; as $V_y \preceq I$, $\mathsf{H}_J \succeq cI$ gives $\lambda_{\max}(\nabla^2 F_{\mathrm{TAP}}(m)) \le -c$.
2. $G = e^{\mathsf{T}}x$, $\|e\| = 1$, in (1.5), p. 6: $\mathrm{Cov}_{\mu_h}(G, x) = \mathrm{Cov}_{\mu_h}(x)e$ and $\mathcal{D}_h(G) = \sum_i e_i^2\,\mathbb{E}\,\mathrm{Var}(x_i \mid x_{-i}) \le 1$, so $s = \|\mathrm{Cov}_{\mu_h}(x)\|$ has $s^2 \le C(s+1)$, $s \le C+1$; Cauchy–Schwarz gives the converse.
3. The two paragraphs after (2.7) in [16, p. 4] against inputs 1–2 of §1.4 (p. 6) and §2.

F11. A weak Poincaré inequality for the zero-field SK heat-bath chain at $\beta < 1/2$

Summary. Davies, Lee, Sandhu and Shi's weak Poincaré paper [14], posted on July 9, 2026, proves that for every $\beta < 1/2$, with high probability over the disorder, the zero-field SK Gibbs measure satisfies a weak Poincaré inequality for the single-site heat-bath (Glauber) chain, with additive error $\epsilon \operatorname{osc}(f)^2$ for every $\epsilon \in (0,1)$ [14, Cor. 1.5, p. 2]. The preprint does not cite this paper anywhere: its account of earlier results in §1.1 (pp. 3–4) omits it, although two later works cited there describe it, Wang's [27, p. 5] and Boban, Li and Oveis Gharan's [28, p. 7]. The earlier result is the inequality of the preprint's main result, Theorem 1.1 (p. 3), with an added error term and only for $\beta < 1/2$ instead of every $\beta < 1$: *weaker result*, since Theorem 1.1 restricted to $\beta < 1/2$ implies it and not conversely. Both proofs observe a spin sample in Gaussian noise up to a large fixed time, prove an inequality for the posterior laws there and transfer it back, an approach the preprint credits to Eldan and to Chen and Eldan in §1.1 (p. 4); in the earlier paper, exceptional localized laws and a weak inequality for the observation law enter as additive $\operatorname{osc}(f)^2$ terms, while the preprint weights paths by f^2 to bound the variance on exceptional ones (p. 4; §2.4, p. 10). The same paper is a technique source in findings F2 and F9.

Correspondence.

	Preprint	Weak Poincaré paper [14]
Measure	μ_0; J has zero diagonal and entries $J_{ij} \sim N(0, j/n)$, $j = \beta^2$ (§1, p. 3)	$\mu_{\beta A}(\sigma) \propto e^{\frac{\beta}{2}\langle \sigma, A\sigma\rangle}$, $A \sim \mathrm{GOE}(n)$ (p. 1): diagonal variance $2/n$, off-diagonal $1/n$ (PHA III [6, p. 1])
Dirichlet form	$\mathcal{D}_0(f) = \sum_i \mathbb{E}_{\mu_0}(f - P_i^0 f)^2$ (1.1), p. 3	Glauber: $\mathcal{E}_\mu(f,f) = \frac{1}{2n}\sum_{x,i} \frac{\mu(x)\mu(x^{\oplus i})}{\mu(x)+\mu(x^{\oplus i})}\big(f(x^{\oplus i}) - f(x)\big)^2$ (p. 6)
Inequality	$\mathrm{Var}_{\mu_0}(f) \le C_\beta \mathcal{D}_0(f)$ for every f (Theorem 1.1, p. 3)	$\mathrm{Var}_\mu(f) \le n e^{C/\epsilon}\mathcal{E}_\mu(f,f) + \epsilon \operatorname{osc}(f)^2$ for every $\epsilon \in (0,1)$: a $(\frac{1}{n}e^{-C/\epsilon}, \epsilon)$-weak Poincaré inequality (Cor. 1.5, p. 2; Def. 2.3, p. 6, with $\mathrm{Err} = \operatorname{osc}^2$ as on p. 7)
Range and disorder	every fixed $\beta < 1$; probability tending to one; C_β depends only on β	$\beta < 1/2$; probability at least $1 - \delta$ for $n \ge N(\beta, \delta)$; $C = C(\beta, \delta)$

How to check.

1. The Glauber Dirichlet form [14, p. 6] against (1.1), p. 3: each pair $\{x, x^{\oplus i}\}$ differing in spin i contributes its mass times the conditional variance of f on it, so $\mathcal{E}_{\mu_0}(f,f) = n^{-1}\mathcal{D}_0(f)$. The diagonal of A adds a constant on $\{\pm 1\}^n$ (p. 3), so $\mu_{\beta A} = \mu_0$ with J the off-diagonal part of βA. Thus Corollary 1.5 reads $\mathrm{Var}_{\mu_0}(f) \le e^{C/\epsilon}\mathcal{D}_0(f) + \epsilon \operatorname{osc}(f)^2$ for $\beta < 1/2$ and $0 < \epsilon < 1$.
2. Theorem 1.1 (p. 3) with $\beta < 1/2$ against [14, Cor. 1.5, p. 2]: the first gives the second with any $C \ge \log C_\beta$, since $e^{C/\epsilon} \ge e^C$ for $\epsilon < 1$; the additive term blocks the converse.
3. The preprint's §1.1 (pp. 3–4) and its bibliography, which do not mention the paper, against [27, p. 5] and [28, p. 7], which describe it.

F12. A positive margin for the Onsager-shifted TAP Hessian by comparison with a norm-weighted Gaussian process

Summary. The proof of Lemma 3.2 (pp. 12–13) names the Sudakov–Fernique inequality and cites Vitale for adding the same deterministic function to both processes (p. 12). The comparison it then makes, of the GOE quadratic form $v^{\mathsf{T}}Wv$ with the norm-weighted linear process $(2/\sqrt{n})\|v\|\langle g, v\rangle$ under a common deterministic shift, is Lemma 4.1 of Celentano, Fan and Mei [11, Lemma 4.1, p. 15], used for the TAP Hessian in their Lemma 4.9 (p. 18; also Lemma 4.12); the preprint does not cite that paper, and cites Celentano, who restates the comparison [18, Prop. 1, p. 6], only for local convexity near AMP iterates (§1.1, p. 4). On p. 13 the preprint takes $v \propto A^{1/2}p$ and the Onsager term as the shift, for each fixed diagonal $0 \le A \le a_+ I$ ($a_+ = 1 + \eta$), and bounds the smallest eigenvalue of $S_A(z; W)$ below through a scalar supremum (3.9); the resulting event feeds Lemmas 3.3–3.5 and 5.5. The shared approach is Celentano, Fan and Mei's: to bound a TAP Hessian below beyond what $\|W\|$ gives, compare its random part with the norm-weighted linear process while the deterministic terms, the Onsager term among them, ride along as a common shift, reducing the smallest eigenvalue to a low-dimensional variational problem; here this gives a margin on the whole range $ja_+^2 < 1$, where $\|W\| \le 2\sqrt{j} + \varepsilon$ alone gives one only when $2\sqrt{j}\, a_+ < 1$. *same technique*: the lemma is used intact, without conditioning and at a fixed diagonal, where the variational problem becomes scalar; Celentano and, not cited, Brennecke, Xu and Yau [17, Prop. 4.5 and Step 1 of its proof, pp. 24–28] build on the same use.

Correspondence.

	Preprint	Celentano–Fan–Mei [11]
Comparison	$\mathbb{E}\sup_{\|p\|=1}\{zX_p - z^2 j\chi_A\|w\|^2\} \le$ $\mathbb{E}\sup_{\|p\|=1}\{zY_p - z^2 j\chi_A\|w\|^2\}$, $X_p = w^{\mathsf{T}}Ww$, $Y_p = 2\sqrt{j/n}\,\|w\|\, g^{\mathsf{T}}w$, $w = A^{1/2}p$ (p. 13)	$\mathbb{E}\sup_{x\in\mathcal{X}}[v(x)^{\mathsf{T}}Wv(x) + f(x)] \le$ $\mathbb{E}\sup_{x\in\mathcal{X}}[\frac{2}{\sqrt{n}}\|v(x)\|\langle g, v(x)\rangle + f(x)]$, $g \sim N(0, I_n)$ (Lemma 4.1, p. 15)
Increments	$\mathbb{E}(Y_p - Y_{p'})^2 - \mathbb{E}(X_p - X_{p'})^2 = \frac{2j}{n}\big[(\|w\|^2 - \|w'\|^2)^2 + 2(\|w\|\|w'\| - w^{\mathsf{T}}w')^2\big] \ge 0$ (p. 13)	$\mathbb{E}\langle v, Wv\rangle\langle v', Wv'\rangle = \frac{2}{n}\langle v, v'\rangle^2$; two inequalities raise $\frac{2}{n}(\|v\|^4 + \|v'\|^4 - 2\langle v, v'\rangle^2)$ to $\frac{2}{n}(2\|v\|^4 + 2\|v'\|^4 - 4\|v\|\|v'\|\langle v, v'\rangle)$; Sudakov–Fernique with common mean f (App. A.4, p. 33)

How to check.

1. The comparison display on p. 13 against [11, Lemma 4.1, p. 15], with $\mathcal{X}$ the unit sphere, $v(p) = (z\sqrt{j})^{1/2}A^{1/2}p$, $f(p) = -z^2 j\chi_A p^{\mathsf{T}}Ap$ and the preprint's W equal to $\sqrt{j}$ times GOE(n): their two processes become zX_p and zY_p.
2. The increment identity on p. 13 against [11, App. A.4, p. 33]: the preprint's bracket is the sum of the two terms the two inequalities drop.
3. $S_A(1; W) = A^{1/2}(A^{-1} + j\chi_A I - W)A^{1/2}$ (§3, p. 11) against the TAP Hessian $H(m)$ of [11, (4.16), p. 16]: with $j = \lambda^2$, $A = \mathrm{diag}(1 - m^2)$ and $\chi_A = 1 - \|m\|^2/n$, $A^{-1} + j\chi_A I - W$ is $H(m)$ without the spike $-\frac{\lambda^2}{n}\mathbf{1}\mathbf{1}^{\mathsf{T}}$ and the term $-\frac{2\lambda^2}{n}mm^{\mathsf{T}}$, and the Onsager term is deterministic in both.

F13. The consistent Gaussian field law: mean equals variance, then a one-variable fixed point

Summary. El Alaoui, Montanari and Sellke (EAMS below) reduce the state evolution of AMP in the planted SK model with Gaussian side observation $tx + \sqrt{t}\,W$ to the fixed point $\gamma_* = \beta^2(1 - \mathrm{mmse}(\gamma_* + t))$, and prove its existence, its uniqueness for $t > 0$ (by concavity), its size of order t and its Lipschitz dependence on t (the last two by comparing fixed-point equations with slope at most β^2) [1, Prop. 4.4, (4.20)–(4.24), pp. 13–15]; for $\beta^2 < 1/3$, Davies, Lee, Sandhu and Shi's Potential Hessian Ascent III (PHA III) proves that every solution of the two-parameter system behind it has equal mean and variance [6, Lemma 5.32, pp. 98–99]. The preprint cites EAMS only in §1.1, p. 4, for their sampling method, and does not cite PHA III; its §4 cites nothing and does not name the Nishimori identity $\mathbb{E}_{N(s,s)} \tanh y = \mathbb{E}_{N(s,s)} \tanh^2 y$ (4.10) on which both steps rest. In Lemma 4.2 (pp. 25–26) it solves the consistency equations $d = t + j\mathbb{E}_P \tanh y$, $s = t + j\mathbb{E}_P \tanh^2 y$ of (4.7) (with $\sigma = 0$) for Gaussian laws $P = N(d, s)$: it forces $d = s$ by the same Lipschitz comparison as PHA III, with constant 1 on $[0, \infty)$ in place of 3, so for every $j < 1$; the remaining equation (4.8) is EAMS's (4.21) under $s_* = t + \gamma_*$, $j = \beta^2$, with the same existence, uniqueness, size and Lipschitz statements and the same comparison for the Lipschitz bound. This is *same technique*: the same scalar system, fixed point and bounds, with the step $d = s$ as in PHA III's Lemma 5.32; the preprint proves uniqueness by a contraction rather than by concavity, and includes $t = 0$. EAMS credit the one-variable reduction and the properties of mmse to Deshpande, Abbe and Montanari [29], who use erasure side information in place of the Gaussian observation.

Correspondence. With $j = \beta^2$, $s_* = t + \gamma_*$ and, by (4.10), $\mathbb{E}_{N(s,s)} \tanh y = 1 - \mathrm{mmse}(s)$ (EAMS), and with $\alpha = t + \beta^2 q^* = s$, $\delta = \beta^2(m^* - q^*) = d - s$ (PHA III):

	Preprint	EAMS [1]; PHA III [6] where named
$d = s$	$H_s(d) = \mathbb{E}_{N(d,s)}[\tanh y - \tanh^2 y]$, 1-Lipschitz on $[0, \infty)$, $H_s(s) = 0$ by (4.10); $\lvert H_s(d)\rvert \le \lvert d - s\rvert = j\lvert H_s(d)\rvert$ (pp. 25–26)	PHA III: $m^* - q^* = \mathbb{E}h(\alpha + \delta + \sqrt{\alpha}G)$, $h = \tanh - \tanh^2$ (5.81), $\mathbb{E}h(\alpha + \sqrt{\alpha}G) = 0$ (5.82) (p. 98); $\lvert h'\rvert \le 3$, $\lvert m^* - q^*\rvert \le 3\lvert\delta\rvert = 3\beta^2\lvert m^* - q^*\rvert$ (p. 99)
Fixed point	(4.8): $s_* = t + j\mathbb{E}_{N(s_*,s_*)} \tanh y$; $q_* = (s_* - t)/j$ (4.9), p. 25	(4.21): $\gamma_* = \beta^2(1 - \mathrm{mmse}(\gamma_* + t))$ (p. 14); $q_* = \gamma_*/\beta^2$, the limit of (2.5) (p. 5)
Size	(4.9): $t \le s_* \le t/(1 - j)$	(4.23); its proof gives $\gamma_* \le \beta^2 t/(1 - \beta^2)$ (p. 15)
Lipschitz	$\lvert s_*(t) - s_*(t')\rvert \le \lvert t - t'\rvert + j\lvert s_*(t) - s_*(t')\rvert$ (p. 26)	Proof of (4.24): $\lvert\gamma_*(t_1) - \gamma_*(t_2)\rvert \le \beta^2\lvert\gamma_*(t_1) - \gamma_*(t_2)\rvert + \beta^2\lvert t_1 - t_2\rvert$ (p. 15)

How to check.

1. The $d = s$ argument (pp. 25–26) against [6, Lemma 5.32 and proof, pp. 98–99]: $H_s(d)$ with (5.81), (4.10) with (5.82), j with $3\beta^2$.
2. (4.8)–(4.9), p. 25 and the Lipschitz comparison on p. 26 against [1, (4.21), (4.23)–(4.24), p. 14] and their proofs (p. 15), with $s_* = t + \gamma_*$ and $j = \beta^2$.

F14. The entropy of the reweighted observation law as the entropy lost along the localization

Summary. The preprint credits Lehec, and Föllmer through Lehec [19], for the drift-energy identity (2.5), which writes the relative entropy of the stopped observation path laws under $f^2\mu_0$ and under μ_0 as half the expected drift energy (§1.1, p. 4, and at the point of use, §2.3, p. 8). It does not cite Chen and Eldan for what this quantity is along the localization: the entropy of f^2 lost, at the rate given by their entropy-decay formula [4, §3.2.3, eq. (27), p. 29]; it cites them only for their variance-retention and Dirichlet-form framework (§1.1, p. 4; §2, p. 7). In §2.3 (pp. 8–9), (2.5) is equivalent to the expectation of (27) with $C_t = I_n$ and f^2 in place of f, integrated up to a stopping time $\tau \le T$, and (2.7) bounds this entropy so as to make the bad paths rare under the reweighted law in (2.8)–(2.9), a step toward Theorem 1.1. *fragment*: the quantity that (2.5) expresses and (2.7) bounds is the entropy loss in (27), but the preprint proves (2.5) by Itô's formula for $\log N_t$, as in the proof of Lehec's Thm 2 [19, pp. 4–5], bounds the mean gap a_t by comparing both means with the TAP root through (1.6) instead of by Chen and Eldan's entropic stability [4, Def. 29, p. 25], and has no counterpart of their Grönwall step and entropy conservation [4, Prop. 39, p. 29].

Correspondence. Chen and Eldan's ν_t (the localized measure) and μ_t (its reweighting) swap the preprint's letters.

	Preprint	Chen–Eldan [4]
Localization	posterior $\mu_{Y(t)}$ of $X \sim \mu_0$, $Y(t) = tX + B(t)$, (1.4), p. 6; innovation $I(t)$, §2.1, p. 7	ν_t from (5) with $C_t = I_n$, the Gaussian channel, driven by B_t (pp. 16–17)
Reweighting	$\nu_t = f^2\mu_{Y(t)}/N_t$, $N_t = \mathbb{E}_t f^2$, $\mathbb{E}_{\mu_0} f^2 = 1$; $\mathbb{P}^f$, $\mathbb{E}^f$: observation law for $X \sim f^2\mu_0$, density N_t, p. 8	$d\mu_t/d\nu_t = f/\nu_t(f)$, $f \ge 0$; $M_t = \nu_t(f)$ (p. 28)
Mass martingale	$dN_t/N_t = a_t^{\mathsf{T}} dI(t)$, $a_t = \mathbb{E}_{\nu_t} x - \mathbb{E}_t x$, p. 8	$dM_t = M_t\langle C_t(\mathrm{b}(\mu_t) - \mathrm{b}(\nu_t)), dB_t\rangle$, b the barycenter (p. 28)
Entropy	$D(\mathbb{P}^f\|_{\mathcal{F}_\tau}\|\mathbb{P}\|_{\mathcal{F}_\tau}) = \mathbb{E}^f \log N_\tau = \frac{1}{2}\mathbb{E}^f \int_0^\tau \|a_t\|^2 dt$, Itô for $\log N_t$, (2.5), p. 8	$d\,\mathrm{Ent}_{\nu_t}[f] = -\frac{1}{2}\nu_t(f)\,\lvert C_t(\mathrm{b}(\mu_t) - \mathrm{b}(\nu_t))\rvert^2 dt + \text{martingale}$, Itô for $M_t \log M_t$, (27), p. 29

Since $\mathbb{E}_t[f^2 \log f^2]$ is a $\mathbb{P}$-martingale and $N_0 = 1$, $\mathbb{E}^f \log N_\tau = \mathrm{Ent}_{\mu_0}(f^2) - \mathbb{E}\,\mathrm{Ent}_{\mu_{Y(\tau)}}(f^2)$ and $\mathbb{E}^f \int_0^\tau \|a_t\|^2 dt = \mathbb{E}\int_0^\tau N_t \|a_t\|^2 dt$, so (2.5) is equivalent to the expectation of (27), with $C_t = I_n$ and f replaced by f^2, integrated from 0 to τ.

How to check.

1. The formula for dN_t/N_t and (2.5) (p. 8) against dM_t and (27) in [4, §3.2.3, pp. 28–29], with $C_t = I_n$ (p. 17) and f^2 in place of f, through the martingale $\mathbb{E}_t[f^2 \log f^2]$.
2. The proof of (2.5) by Itô's formula for $\log N_t$ (p. 8) against the proof of [19, Thm 2, pp. 4–5].
3. The bound on $\|a_t\|/\sqrt{n}$ from (1.6) and the bound (2.7) (p. 8) against Def. 29 and Lemma 31 (p. 25) and [4, Prop. 39, p. 29].

F15. Comparing tanh at two fields through the Stein operator of a two-Gaussian mixture

Summary. Chatterjee [20, §1.8 and Lemma 1.3, pp. 7–8] develops "Stein's method for mixture Gaussians". For the law $M(a,b,\mu,\sigma^2)$, with density proportional to $\cosh(ax+b)e^{-(x-\mu)^2/2\sigma^2}$, a two-Gaussian mixture under which $\tanh(ax+b)$ has mean $\tanh(a\mu+b)$, the paper gives a characterizing operator and bounded solutions of its Stein equation, and uses them, centred at the TAP field, to explain and prove the TAP equations. The preprint does not cite this paper, and its compiled text does not mention Stein's method. For $a>0$, its scalar identity (6.9), p. 45 is $-a$ times Chatterjee's Stein equation with $u=\tanh$, for the parameters $a=1$, $b=0$, $\mu=r_0$ and σ^2 equal to the preprint's a, and its kernel Φ (6.8) is $-1/a$ times the bounded solution there. The shared approach, and Chatterjee's insight (Chatterjee's §1.4, p. 4, is titled "Onsager correction and mixture Gaussians"), is to write tanh at a field minus tanh at a centre as this operator, whose multiplicative part is the field minus the centre minus a coefficient times tanh of the field, applied to a bounded function, leaving terms that integration by parts controls; the preprint does this in each induction step of its weak residual rule (Lemma 6.5, pp. 53–54) and, centred at the root of the field equation, in (6.12)–(6.13), pp. 45–46, and (7.3), p. 57. *fragment*: the preprint uses only the operator and its tanh solution, as an exact identity whose a is a difference of Onsager coefficients of either sign, without the mixture law, Lemma 1.3 for general u, or Gaussian integration by parts in the disorder.

Correspondence.

	Preprint	Chatterjee [20]
Operator	$(z-r_0-a\tanh z)-a\partial_z$, $z,r_0,a\in\mathbb{R}$ (6.9)	$Tf=f'-\big(\frac{x-\mu}{\sigma^2}-a\tanh(ax+b)\big)f$ for $M(a,b,\mu,\sigma^2)$, $\sigma^2>0$ (12), p. 7
Right side	$\tanh z-\tanh r_0$ (6.9)	$u-\int u\,\psi$ (15), equal to $\tanh x-\tanh\mu$ for $u=\tanh$ by (14), p. 7
Bounded solution	Φ (6.8); $0<\Phi\le e^{\lvert a\rvert/2}$; for a in a bounded interval, fixed-order derivatives at most $C\Phi$, uniformly in z,r_0 (p. 45)	$f=\rho^{-1}\int_{-\infty}^{x}\rho\,(u-r(\mu))$, display before (54), p. 25; $\lvert f\rvert,\lvert\partial_x f\rvert,\lvert\partial_\mu f\rvert\le C(a,\sigma)\lVert u\rVert_\infty$ (Lemma 1.3, p. 8)
Centre	$r_0=h_i^{p+1}$ (Lemma 6.5, p. 54); the root r_i (6.12)	the TAP field r_1 of (10), p. 5; law $M(\beta,h,r_1,1-q)$ (pp. 20–22)

How to check.

1. (6.9), p. 45 against T and (14)–(15) in [20, §1.8, p. 7], with Chatterjee's $a=1$, $b=0$, $\mu=r_0$ and σ^2 equal to the preprint's $a>0$: $-\sigma^2T$ is the preprint's operator, and for $u=\tanh$ the right side of (15) is $\tanh z-\tanh r_0$.
2. (6.8) and the bounds below it, p. 45 against the display before (54) in [20, §3.6, p. 25] with $u=\tanh$, and Lemma 1.3 (p. 8): for $a>0$, $-a\Phi$ is a bounded solution of (15), and the other solutions differ by multiples of $1/\rho=e^{(x-\mu)^2/2\sigma^2}/\cosh x$, which is unbounded, so $\Phi=-f/a$.
3. (6.12)–(6.13), pp. 45–46, and (7.3), p. 57 against the centring at the TAP field r_1 in $\langle(\ell_1-r_1-\beta(1-q)\tanh(\beta\ell_1+h))f\rangle-(1-q)\langle f_1\rangle=-(1-q)\langle Tf\rangle$, with ℓ_1 the local field [20, p. 22].

F16. Comparing a posterior mean with the TAP root via a late iterate

Summary. El Alaoui, Montanari and Sellke, for $\beta < 1/2$, place the mean $m(A, y)$ of the SK Gibbs measure at a stochastic-localization field within $\rho\sqrt{tn}$ of a stationary point m_* of the TAP free energy through a late AMP iterate: the mean is close to it, its small increments make it an approximate TAP solution, and local strong convexity puts m_* near it [1, Lemma 4.10(2), p. 20; proof, p. 40]. The preprint cites that paper only for its sampling method, in §1.1 (p. 4), and cites nothing in §6.4 (p. 45) or Proposition 7.3 (p. 58), where it compares the mean of $\nu = f^2\mu_{h_0}/N$, $N = \mathbb{E}_{\mu_{h_0}} f^2$, with t_* through a late iterate of its recursion; both papers need the comparison for the drift of a localization process, in [1, Lemma 4.14, p. 23] and in §2.3 (p. 8). It cites Bolthausen for its recursion; Bolthausen's iterates converge [26, Thm 2.1, p. 4] but are not compared with the Gibbs measure [26, p. 2]. *fragment*: the preprint shares the outline and replaces both substantive links: for a recursion started at a Gibbs sample and under f^2 reweighting, its mean–iterate link is an $O(1)$ bound in expectation (Lemma 6.5) built by induction on the exact conditional-spin identity (6.31), p. 53, with small increments from pigeonhole (Lemma 7.1); its root link is the global bound of Lemma 5.7 (p. 40), not convexity near an iterate. The outline appears earlier in Celentano, Fan and Mei for $\mathbb{Z}_2$-synchronization, from a Bayes-optimal late AMP iterate to a TAP local minimizer by local strong convexity [11, Thm 2.1(a), p. 8; p. 18] (not cited); Celentano's convexity theorem, which the preprint cites in §1.1, extends El Alaoui, Montanari and Sellke's link to all $\beta < 1$ [18, Thm 2, pp. 10–11; p. 4].

Correspondence.

	Preprint	El Alaoui–Montanari–Sellke
Mean ≈ iterate	$\lvert\mathbb{E}_\nu[\widehat{C}_k(x - m^k)^\mathsf{T} e]\rvert \le C_\delta(1 + \mathcal{D}_{h_0}(f)/N)$, $\lVert e\rVert \le 1$, cutoffs $\widehat{C}_k$ of Lemma 7.1 (proof of Prop. 7.3, p. 58)	$\lVert m(A, y) - \hat{m}^k\rVert/\lVert m(A, y)\rVert \to 0$, AMP iterate $\hat{m}^k = \tanh z^k$ of (4.15) (Prop. 4.8, (4.37), p. 17)
Iterate ≈ approx. root	$F_{J,h_0}(h^k) =$ $JR_k - jb_{k-1}(R_{k-1} + R_k) + j(b_k - b_{k-1})m^k$, $\lvert b_k - b_{k-1}\rvert \le 2\lVert R_k\rVert/\sqrt{n}$ (p. 45)	$\lVert z^{k+1} - z^k\rVert/\lVert z^k\rVert \to 0$ (4.38) gives $\lVert\nabla\mathcal{F}_{\mathrm{TAP}}(\hat{m}^k; y, q)\rVert/\sqrt{tn} \to 0$ (Lemma 4.11, p. 21)
Approx. root ≈ root	$\lVert y - r\rVert \le C\lVert F_{J,h_0}(y)\rVert$ if $\lVert F_{J,h_0}(y)\rVert < r_0\sqrt{n}$, given $\mathsf{H}_J(y, V_y) \succeq cI$ on that whole region ((5.29), p. 40)	convexity for $\beta \le 1/2 - c$ (Lemma 4.12, p. 22) and Lemma A.6 (p. 35): $\lVert\hat{m}^{K_{\mathsf{AMP}}} - m_*\rVert \le \rho\sqrt{tn}/2$ (p. 40)
Conclusion	$\lVert\mathbb{E}_\nu x - t_*\rVert/\sqrt{n} \le$ $C\delta + C_\delta n^{-1/2}(1 + \mathcal{D}_{h_0}(f)/N)$ ((7.6), p. 58)	$\lVert m(A, y) - m_*(A, y)\rVert_2 \le \rho\sqrt{tn}$ (Lemma 4.10(2), p. 20)

How to check.

1. The proof of Proposition 7.3, p. 58 against the proof of [1, Lemma 4.10, Part 2, p. 40] and the argument for [11, Thm 2.1(a), p. 18]: mean to late iterate, iterate to root, triangle.
2. The $F_{J,h_0}(h^k)$ identity after (6.11), p. 45 against the proof of [1, Lemma 4.11, p. 21], with $\beta A \to J$, $\beta^2 \to j$, $y \to h_0$, $z^k \to h^k$, $q = \lVert\hat{m}^k\rVert^2/n$.

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